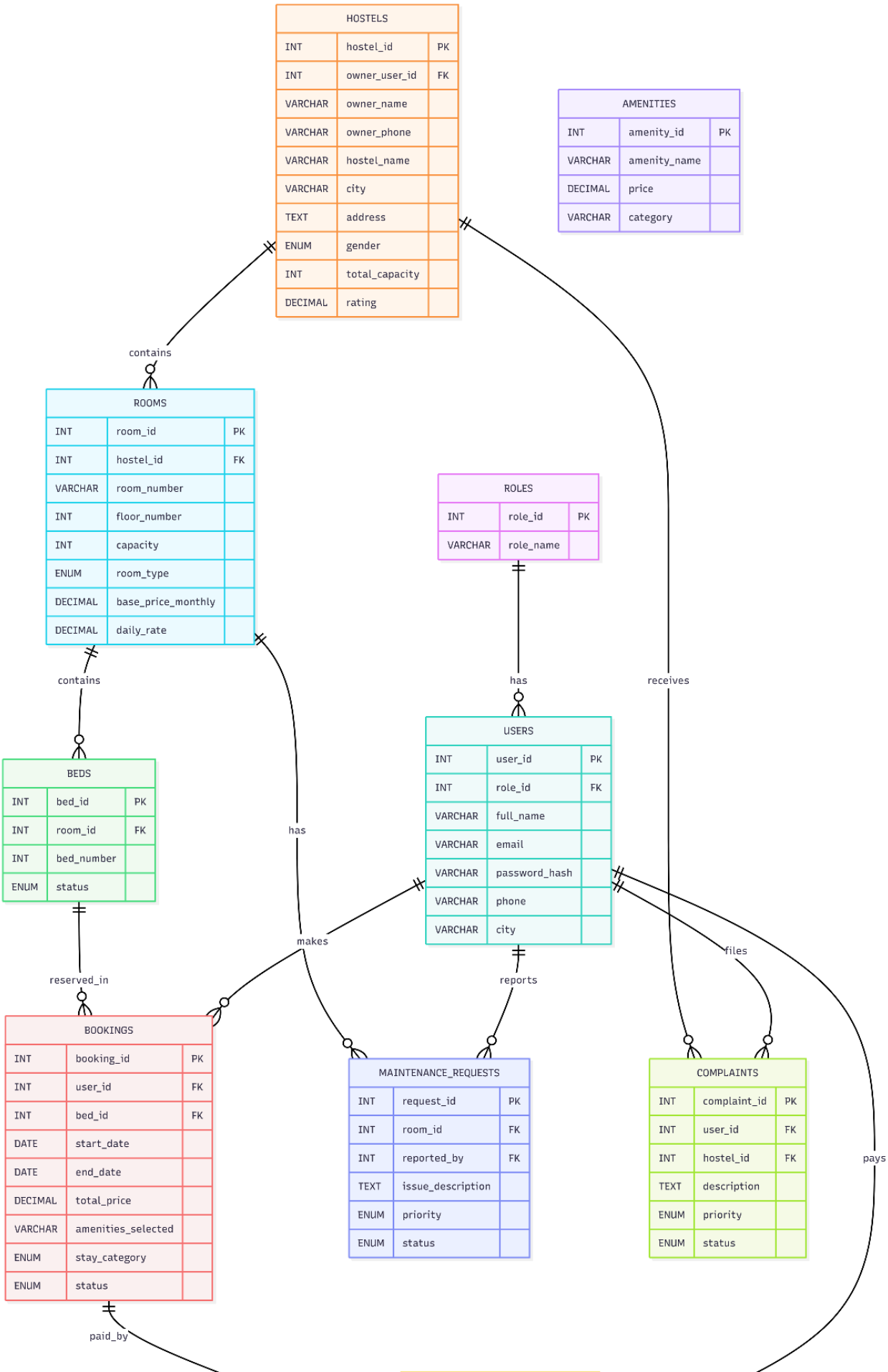
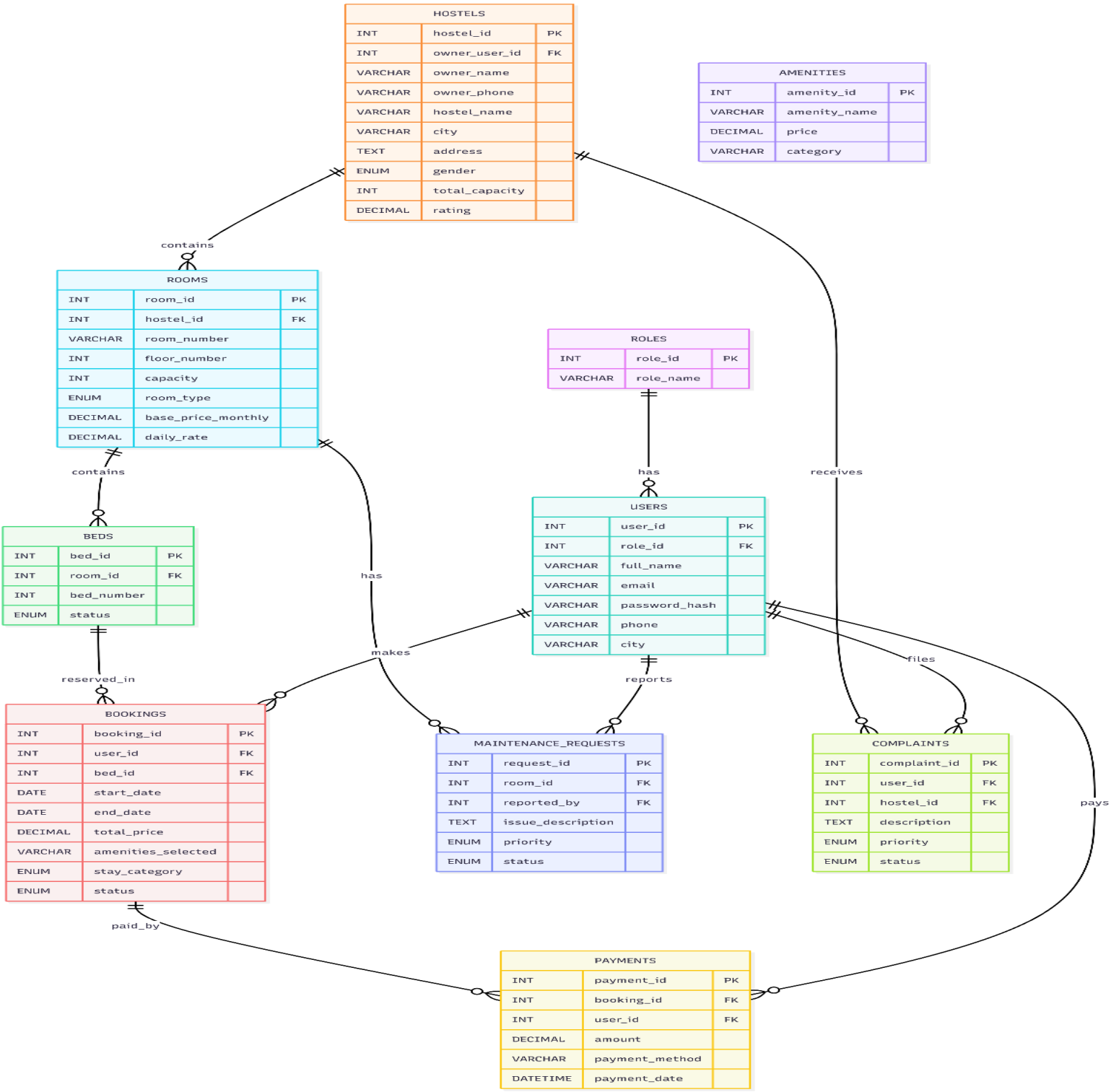
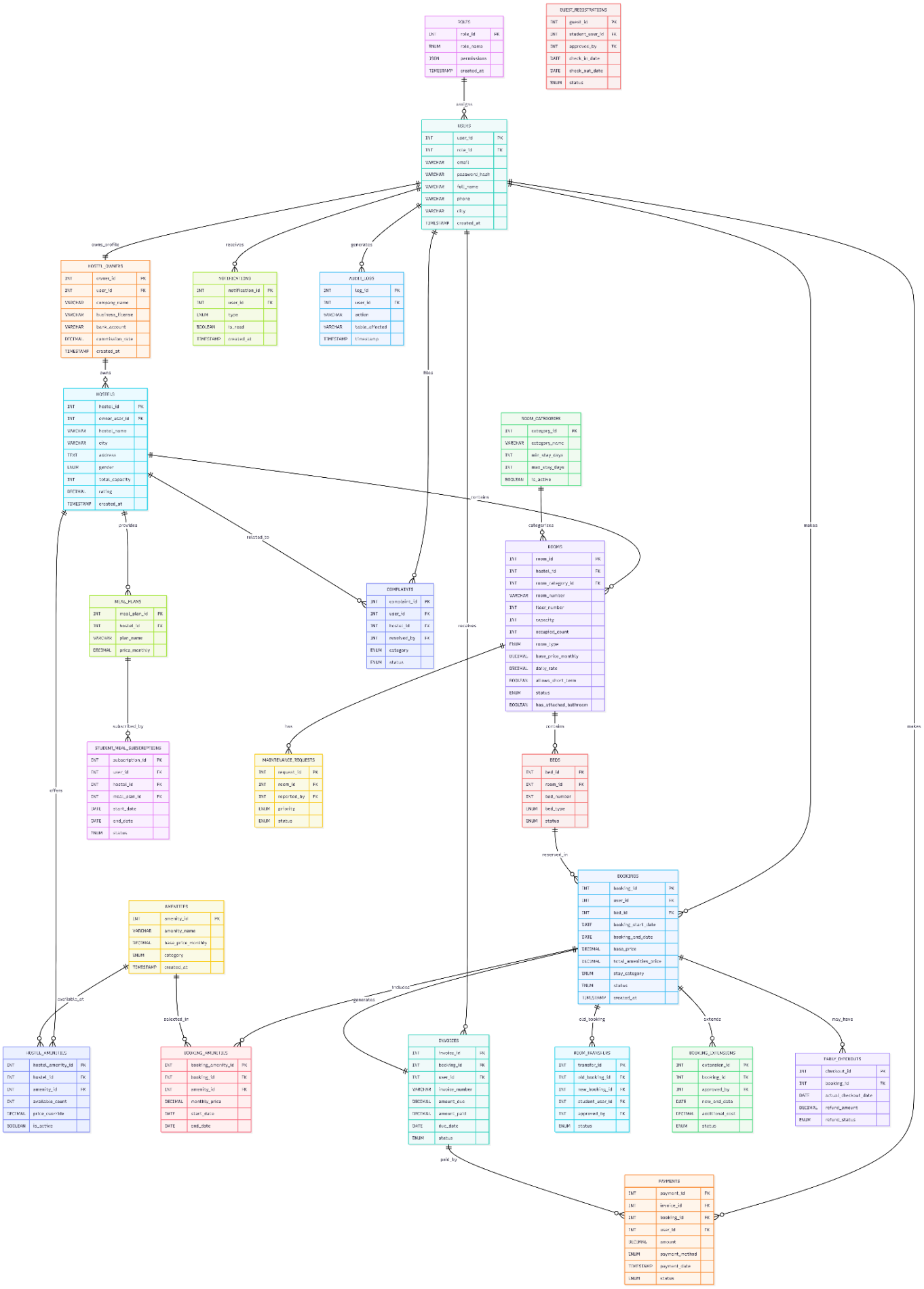
Entity Relationship Diagram

BSCS24138
BSCS24028
1NF:


HOSTELS
INT hostel_id PK
INT owner_user_id FK
VARCHAR owner_name
VARCHAR owner_phone
VARCHAR hostel_name
VARCHAR city
TEXT address
ENUM gender
INT total_capacity
DECIMAL rating
AMENITIES
INT amenity_id PK
VARCHAR amenity_name
DECIMAL price
VARCHAR category
contains
ROOMS
INT room_id PK
INT hostel_id FK
VARCHAR room_number
INT floor_number
INT capacity
ENUM room_type
DECIMAL base_price_monthly
DECIMAL daily_rate
ROLES
INT role_id PK
VARCHAR role_name
receives
has
contains
has
USERS
INT user_id PK
INT role_id FK
VARCHAR full_name
VARCHAR email
VARCHAR password_hash
VARCHAR phone
VARCHAR city
BEDS
INT bed_id PK
INT room_id FK
INT bed_number
ENUM status
makes
files
reserved_in
reports
BOOKINGS
INT booking_id PK
INT user_id FK
INT bed_id FK
DATE start_date
DATE end_date
DECIMAL total_price
VARCHAR amenities_selected
ENUM stay_category
ENUM status
MAINTENANCE_REQUESTS
INT request_id PK
INT room_id FK
INT reported_by FK
TEXT issue_description
ENUM priority
ENUM status
COMPLAINTS
INT complaint_id PK
INT user_id FK
INT hostel_id FK
TEXT description
ENUM priority
ENUM status
pays
paid_by


HOSTELS
INT hostel_id PK
INT owner_user_id FK
VARCHAR owner_name
VARCHAR owner_phone
VARCHAR hostel_name
VARCHAR city
TEXT address
ENUM gender
INT total_capacity
DECIMAL rating
AMENITIES
INT amenity_id PK
VARCHAR amenity_name
DECIMAL price
VARCHAR category
contains
ROOMS
INT room_id PK
INT hostel_id FK
VARCHAR room_number
INT floor_number
INT capacity
ENUM room_type
DECIMAL base_price_monthly
DECIMAL daily_rate
ROLES
INT role_id PK
VARCHAR role_name
has
receives
contains
BEDS
INT bed_id PK
INT room_id FK
INT bed_number
ENUM status
has
USERS
INT user_id PK
INT role_id FK
VARCHAR full_name
VARCHAR email
VARCHAR password_hash
VARCHAR phone
VARCHAR city
makes
files
reserved_in
reports
BOOKINGS
INT booking_id PK
INT user_id FK
INT bed_id FK
DATE start_date
DATE end_date
DECIMAL total_price
VARCHAR amenities_selected
ENUM stay_category
ENUM status
MAINTENANCE_REQUESTS
INT request_id PK
INT room_id FK
INT reported_by FK
TEXT issue_description
ENUM priority
ENUM status
COMPLAINTS
INT complaint_id PK
INT user_id FK
INT hostel_id FK
TEXT description
ENUM priority
ENUM status
pays
paid_by
PAYMENTS
INT payment_id PK
INT booking_id FK
INT user_id FK
DECIMAL amount
VARCHAR payment_method
DATETIME payment_date

3NF


ROLES
INT role_id PK
ENUM role_name
JSON permissions
TIMESTAMP created_at
GUEST_REGISTRATIONS
INT guest_id PK
INT student_user_id FK
INT approved_by FK
DATE check_in_date
DATE check_out_date
ENUM status
assigns
USERS
INT user_id PK
INT role_id FK
VARCHAR email
VARCHAR password_hash
VARCHAR full_name
VARCHAR phone
VARCHAR city
TIMESTAMP created_at
owns_profile
receives
generates
files
receives
makes
makes
HOSTEL_OWNERS
INT owner_id PK
INT user_id FK
VARCHAR company_name
VARCHAR business_license
VARCHAR bank_account
DECIMAL commission_rate
TIMESTAMP created_at
NOTIFICATIONS
INT notification_id PK
INT user_id FK
ENUM type
BOOLEAN is_read
TIMESTAMP created_at
AUDIT_LOGS
INT log_id PK
INT user_id FK
VARCHAR action
VARCHAR table_affected
TIMESTAMP timestamp
owns
HOSTELS
INT hostel_id PK
INT owner_user_id FK
VARCHAR hostel_name
VARCHAR city
TEXT address
ENUM gender
INT total_capacity
DECIMAL rating
TIMESTAMP created_at
ROOM_CATEGORIES
INT category_id PK
VARCHAR category_name
INT min_stay_days
INT max_stay_days
BOOLEAN is_active
contains
provides
related_to
categorizes
MEAL_PLANS
INT meal_plan_id PK
INT hostel_id FK
VARCHAR plan_name
DECIMAL price_monthly
COMPLAINTS
INT complaint_id PK
INT user_id FK
INT hostel_id FK
INT resolver_by FK
ENUM category
ENUM status
receives
ROOMS
INT room_id PK
INT hostel_id FK
INT room_category_id FK
VARCHAR room_number
INT floor_number
INT capacity
INT occupied_count
ENUM room_type
DECIMAL base_price_monthly
DECIMAL daily_rate
BOOLEAN allows_short_term
ENUM status
BOOLEAN has_attached_bathroom
subscribed_by
has
contains
makes
STUDENT_MEAL_SUBSCRIPTIONS
INT subscription_id PK
INT user_id FK
INT hostel_id FK
INT meal_plan_id FK
DATE start_date
DATE end_date
ENUM status
offers
MAINTENANCE_REQUESTS
INT request_id PK
INT room_id FK
INT reported_by FK
ENUM priority
ENUM status
BEDS
INT bed_id PK
INT room_id FK
INT bed_number
ENUM bed_type
ENUM status
reserved_in
BOOKINGS
INT booking_id PK
INT user_id FK
INT bed_id FK
DATE booking_start_date
DATE booking_end_date
DECIMAL base_price
DECIMAL total_amenities_price
ENUM stay_category
ENUM status
TIMESTAMP created_at
AMENITIES
INT amenity_id PK
VARCHAR amenity_name
DECIMAL base_price_monthly
ENUM category
TIMESTAMP created_at
includes
generates
available_at
selected_in
old_booking
extends
may_have
HOSTEL_AMENITIES
INT hostel_amenity_id PK
INT hostel_id FK
INT amenity_id FK
INT available_count
DECIMAL price_override
BOOLEAN is_active
BOOKING_AMENITIES
INT booking_amenity_id PK
INT booking_id FK
INT amenity_id FK
DECIMAL monthly_price
DATE start_date
DATE end_date
INVOICES
INT invoice_id PK
INT booking_id FK
INT user_id FK
VARCHAR invoice_number
DECIMAL amount_due
DECIMAL amount_paid
DATE due_date
ENUM status
ROOM_TRANSFERS
INT transfer_id PK
INT old_booking_id FK
INT new_booking_id FK
INT student_user_id FK
INT approved_by FK
ENUM status
BOOKING_EXTENSIONS
INT extension_id PK
INT booking_id FK
INT approved_by FK
DATE new_end_date
DECIMAL additional_cost
ENUM status
EARLY_CHECKOUTS
INT checkout_id PK
INT booking_id FK
DATE actual_checkout_date
DECIMAL refund_amount
ENUM refund_status
paid_by
PAYMENTS
INT payment_id PK
INT invoice_id FK
INT booking_id FK
INT user_id FK
DECIMAL amount
ENUM payment_method
TIMESTAMP payment_date
ENUM status